

# Supplementary Information on “Restoration of motion-blurred, high-resolution Mars Express SRC images of Phobos”

Ryodo Hemmi and Hiroshi Kikuchi

## 1. Details of denoising

The MEx SRC level 3 data products are already calibrated as the MEx team has officially reduced most of the pepper noise (anomalously dark or NoData pixels recorded in original images) left in the level 2 products. “Pepper noise,” referring to spurious dark or bright pixels caused by sensor readout artifacts, is partially removed in level 3 data but requires additional filtering for accurate deconvolution. We regarded these residual pepper pixels as NULL pixels using the ISIS “specpix”, which would cause problematic errors when subsequent processing in frequency-domain. Also, each PDS label contains the parameter “LINE\_FIRST\_PIXEL = 3”, which is intended to indicate that the first two lines do not contain valid data. However, this assumption does not always hold because several of the individual cubes have data lines within the first three lines. Thus, we marked the pixels having DNs less than 200 within the first three lines as “Low Representation Saturation” (LRS; LRS pixels with  $DN \leq -200$  correspond to invalid or saturated detector readings and were empirically identified through histogram analysis of dark frames). And then we completely removed them as well as dark pixels (75 NULL pixels per cube) by replacing the mean of their surrounding pixels by running ISIS “noisefilter” (third and fourth boxes in Main Text Figure 1). We repeated this noise filtering process until no LRS pixels remained in the resulting cube (twice for h5851\_0003\_sr3 and three times for h5851\_0004\_sr3). Moreover, any pixel whose raw DN value was  $\leq -200$  was considered anomalously dark and substituted with the average DN of its eight neighboring pixels. As exemplified in Figure S1, the NULL and dark pixels are removed from all SRC cubes.

Thresholds such as  $DN < 200$  for LRS pixels were determined empirically by analyzing pixel intensity histograms across multiple images and verifying that all low-DN pixels corresponded to non-illuminated detector rows. To confirm that denoising did not bias deconvolution, we processed test images with and without denoising (Figure S2). No artificial ringing or spurious features were introduced, indicating that denoising only removes non-informative noise while preserving real surface details.

Our full Python implementation, including SPICE geometry handling and Wiener filtering, will be publicly released at <https://github.com/ryodo-hemmi/public> and <https://doi.org/10.5281/zenodo.17119672>. Sample Shell script codes of image processing by means of ISIS commands are as follows.

---

```
ls h5851/*_00{02..04}_sr3.img | sed -e "s#.img##" | tee h5851.lis
ls h7926/*_0011_sr3.img | sed -e "s#.img##" | tee h7926.lis
ls h8974/*_00{02..08}_sr3.img | sed -e "s#.img##" | tee h8974.lis
ls he776/*_00{02..04}_sr3.img | sed -e "s#.img##" | tee he776.lis
cat h5851.lis h7926.lis h8974.lis he776.lis | tee all.lis
hrsc2isis from=¥$1.img to=¥$1.cub -batch=all.lis
spiceinit from=¥$1.cub -batch=all.lis cksmithed=true spksmithed=true extra=~/src/spice/DATA/SPK/MEX_STRUCT_V01.BSP ¥
    shape=user model=~/src/ernst2023/phobos_g_018m_spc_0000n00000_v002.bds
fx f1=¥$1.cub to=¥$1.fx.cub+16-bit -batch=all.lis
    equation="¥(f1*(line>3))+(f1*(f1>=200)*(line<=3))+(-32767*(f1<200)*(line<=3))"
specpix from=¥$1.fx.cub to=¥$1.sp.cub -batch=all.lis nullmin=-1 nullmax=-1
noisefilter from=¥$1.sp.cub to=¥$1.nf.cub -batch=all.lis ¥
    tolmin=100000 tolmax=100000 samples=3 lines=3 low=200 ¥
    replace=average nullisnoise=true lrsisnoise=true
cp h5851/h5851_0003_sr3.nf.cub h5851/h5851_0003_sr3.nf1.cub
cp h5851/h5851_0004_sr3.nf.cub h5851/h5851_0004_sr3.nf1.cub
noisefilter from=h5851/h5851_0003_sr3.nf1.cub to=h5851/h5851_0003_sr3.nf.cub ¥
    tolmin=100000 tolmax=100000 samples=3 lines=3 low=200 ¥
    replace=average nullisnoise=true
noisefilter from=h5851/h5851_0004_sr3.nf1.cub to=h5851/h5851_0004_sr3.nf2.cub ¥
    tolmin=100000 tolmax=100000 samples=3 lines=3 low=200 ¥
    replace=average nullisnoise=true
```

```

noisefilter from=h5851/h5851_0004_sr3.nf2.cub to=h5851/h5851_0004_sr3.nf.cub ¥
    tolmin=100000 tolmax=100000 samples=3 lines=3 low=200 ¥
    replace=average nullisnoise=true
while read line; do set $line; python raster_fill.py $1.nf.cub -o $1.cal.cub -t -200; done < all.lis
while read line; do set $line; python deblur_cub.py $1.cal.cub -log -v -se -sn 16 -t -ev; done < all.lis

```

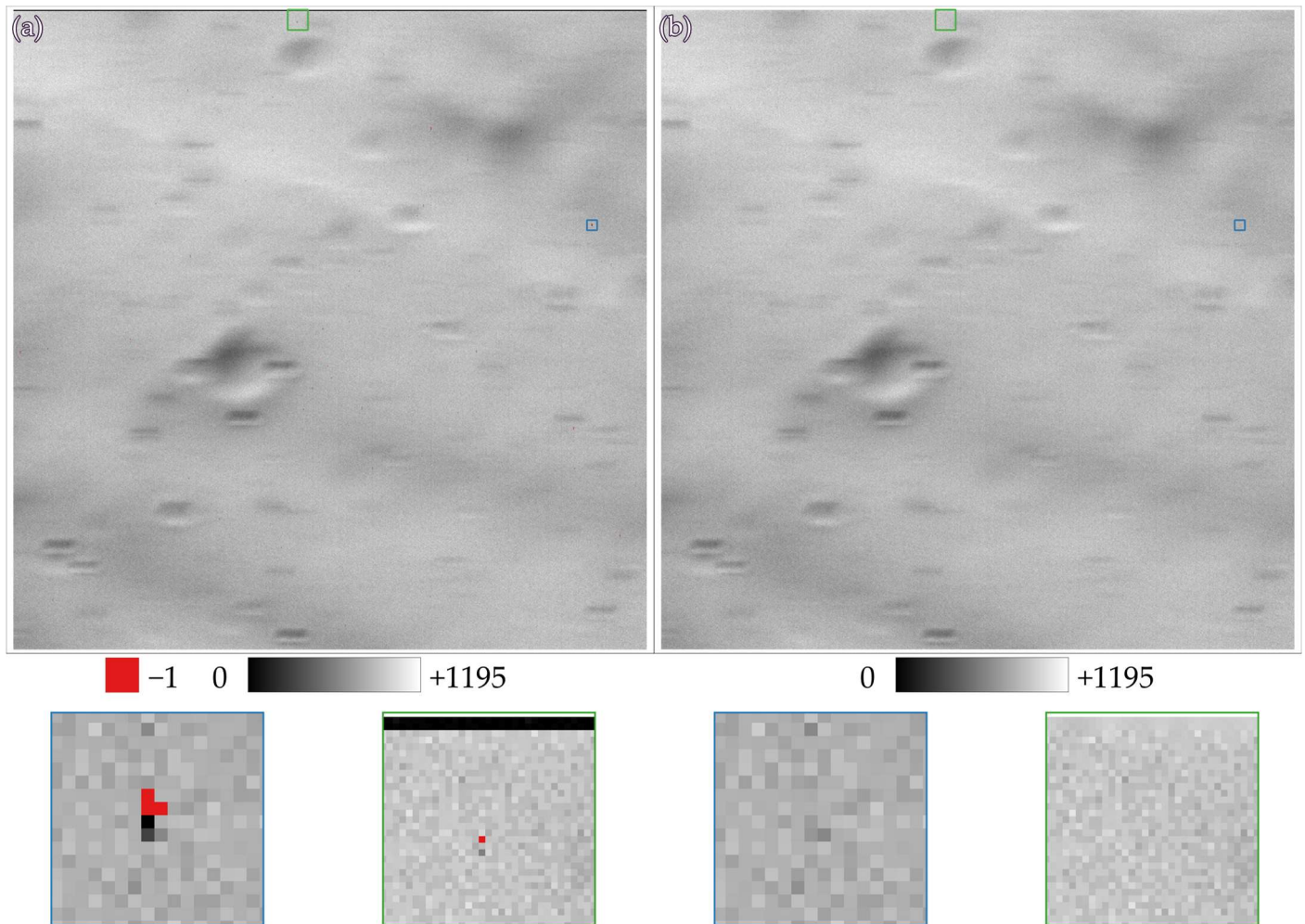


Figure S1. An SRC image example (a) before and (b) after denoising processing. Red pixels in the panel (a) represent the pepper pixels having NULL values (DN = -1; A close-up view of a digitized PDF version is recommended for readers). The first two dark lines of LRS pixels in (a) are also cleaned after this processing (see panel b). For comparison with the pre-processed image in (a), the denoised image in (b) is shown in a range from 0 to +1195, while its actual range is from +334 to +1195. Closeup views of specified pixels (DN = 0 and DN = -1) before and after denoising (on the left and right figures, respectively) are shown at the bottom. Locations are shown as blue and green boxes in the top figures. SRC image ID is h5851\_0003\_sr3.

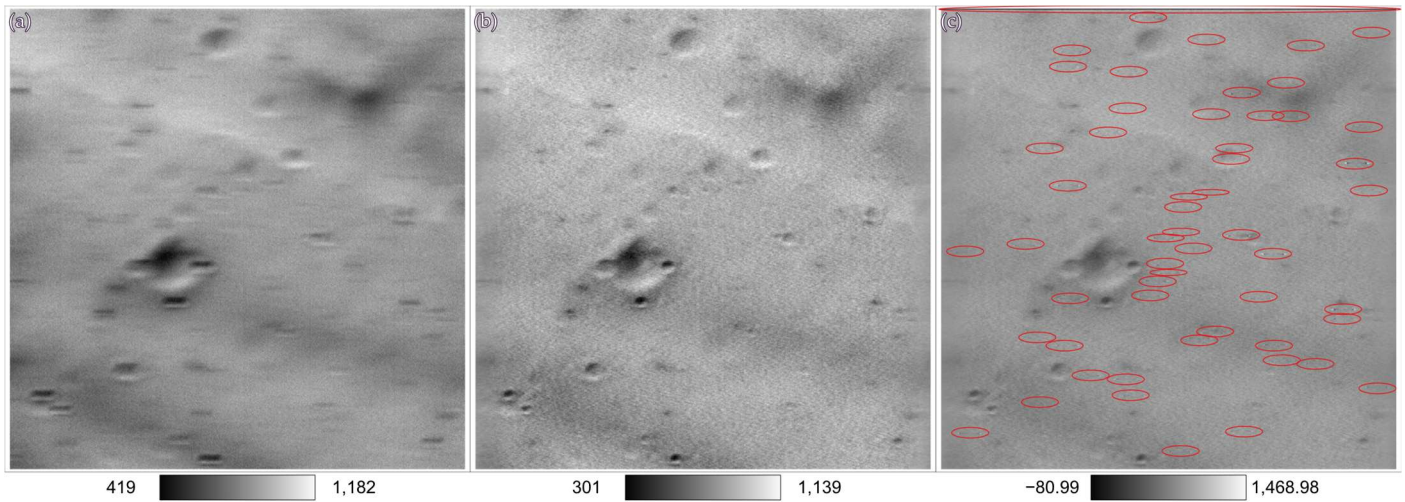


Figure S2. Impacts of denoising processing after restoration. (a) a denoised image before restoration, (b) Restoration of a denoised image (c) Restoration of a non-denoised image. SRC image. Pepper noises and LRS pixels induce severe artifacts with anomalously enhanced bright/dark pixels (marked by red ellipses) after restoration. SRC image ID is h5851\_0003\_sr3.

## 2. Details of PSF estimation from SRC observation geometries

After collecting values regarding the IMG/cube labels "START\_TIME" (UTC1), "STOP\_TIME" (UTC2), and "EXPOSURE\_DURATION" (exposure) with the USGS Abstraction Layer for Ephemerides (ALE) library [1-3], we first performed calculation of "high-accuracy" observation start and stop timings in forms of ephemeris time (ET) with micro-second precisions (ET'1 and ET'2,) based on:

$$ET'_1 = (ET_1 + ET_2)/2 - exposure/2 = t_1, \quad (S1)$$

$$ET'_2 = (ET_1 + ET_2)/2 + exposure/2 = t_2, \quad (S2)$$

where ET1 and ET2 are the ET expressions of UTC1 and UTC2, respectively.

Conversions from ET to UTC and vice versa were implemented by uses of Spicypy et2utc and utc2et, respectively. The resulted UTC'1 and UTC'2 (UTC expressions of the resulted ET'1 and ET'2) about the selected SRC images are listed in Table S2.

For PSF estimation, we started calculating the SRC boresight intersections with the Phobos surface DSK in the Phobos' body-fixed reference frame "IAU\_PHOBOS". Using Spicypy [4] sincpt function with the aberration correction option of "CN+S" (both light time and stellar correction) gives two positional vectors at ET1' simultaneously: one from observer (obs; the origin of SRC boresight, "MEX\_HRSC" whose SPICE body code is -41200) to the surface-intersection point (spt) ( $\vec{r}_{obs \rightarrow spt}(t_1)$  in Figure S3) and another from Phobos center (bod) to the same spt ( $\vec{r}_{bod \rightarrow spt}(t_1)$  in Figure S3). Taking their vector difference gives  $\vec{r}_{obs \rightarrow bod}(t_1)$  (as shown in Figure S3):

$$\vec{r}_{obs \rightarrow bod}(t_1) = \vec{r}_{obs \rightarrow spt}(t_1) - \vec{r}_{bod \rightarrow spt}(t_1) \quad (S3)$$

Similarly, using Spicypy spkpos function with the "CN+S" aberration correction, we obtain a positional vector at ET2' from obs to bod ( $\vec{r}_{obs \rightarrow bod}(t_2)$  in Figure S3) as well. Then, using it with the  $\vec{r}_{obs \rightarrow bod}(t_1)$  resulted from equation (S3), we can calculate the vector directing from obs at ET2' to obs at ET1':

$$\vec{r}_{obs \rightarrow obs}(t_2) = \vec{r}_{obs \rightarrow bod}(t_2) - \vec{r}_{obs \rightarrow bod}(t_1) \quad (S4)$$

Using  $\vec{r}_{obs \rightarrow obs}(t_2)$  derived from equation (S4) and  $\vec{r}_{obs \rightarrow spt}(t_1)$ , we finally gain the positional vector at ET2' from obs to spt ( $\vec{r}_{obs \rightarrow spt}(t_2)$  in Figure S3):

$$\vec{r}_{obs \rightarrow spt}(t_2) = \vec{r}_{obs \rightarrow obs}(t_2) + \vec{r}_{obs \rightarrow spt}(t_1) \quad (S5)$$

The two resulting position vectors from obs to spt at ET1' and ET2',  $\vec{r}_{obs \rightarrow spt}(t_1)$  and  $\vec{r}_{obs \rightarrow spt}(t_2)$ , are expressed in the Phobos-body fixed frame "IAU\_PHOBOS" (SPICE FK ID: 10021). We thus translated these vectors into the ones in the instrument frame "MEX\_HRSC\_SRC" (SPICE FK ID: -41220) by multiplying the frame-to-frame ("IAU\_PHOBOS" to "MEX\_HRSC\_SRC") transformation matrices at ET1' and ET2' that were generated by Spicypy function pxform. The SPICE kernels used for frame transformations are provided by ESA and JPL and yield spacecraft positional accuracies on the order of **tens of meters**, far below the image spatial resolution (~0.5–3 m/pixel). Therefore, transformation errors have negligible impact on PSF estimation accuracy.

Using two obs-to-spt vectors expressed in the "MEX\_HRSC\_SRC" frame, we derived image coordinates of the two corresponding points on the SRC image plane:

$$X_1 = (focal\ length/pixel\ size_x) (x_1/z_1), \quad (S6)$$

$$X_2 = (focal\ length/pixel\ size_x) (x_2/z_2), \quad (S7)$$

$$Y_1 = (focal\ length/pixel\ size_y) (x_1/z_1), \quad (S8)$$

$$Y_2 = (focal\ length/pixel\ size_y) (y_2/z_2), \quad (S9)$$

where  $(X1, Y1)$  and  $(X2, Y2)$  are respectively the coordinates of the projected point at ET1' and ET2',  $(x1, y1, z1)$  and  $(x2, y2, z2)$  are respectively the components of  $\vec{r}_{obs \rightarrow spt}(t_1)$  and  $\vec{r}_{obs \rightarrow spt}(t_2)$  expressed in the "MEX\_HRSC\_SRC" frame; focal length is  $9.8476 \times 10^5 \mu\text{m}$  ( $= 984.76 \text{ mm}$  [5]), pixel sizex is  $9.0 \mu\text{m/pixel}$ , and pixel sizey is  $9.0 \mu\text{m/pixel}$  [6], each of which is stored in SPICE IK MEX\_HRSC\_V09.TI.

Finally, we obtained the shift of the projected points during exposure on the SRC image plane by calculating their differences:

$$shift_x = X_2 - X_1 \quad (S10)$$

$$shift_y = Y_2 - Y_1, \quad (S11)$$

The obtained PSF geometries are summarized in Table S3. By using longer exposures—values matching the SRC observation durations—our pipeline extracted extended PSFs, which is consistent with the initial expectation of this study. Most of the vertical components of the shifts are close to zero relative to horizontal components, so that the resulting PSF can be approximated to be a horizontally flat line (See Figure 2 for a typical case).

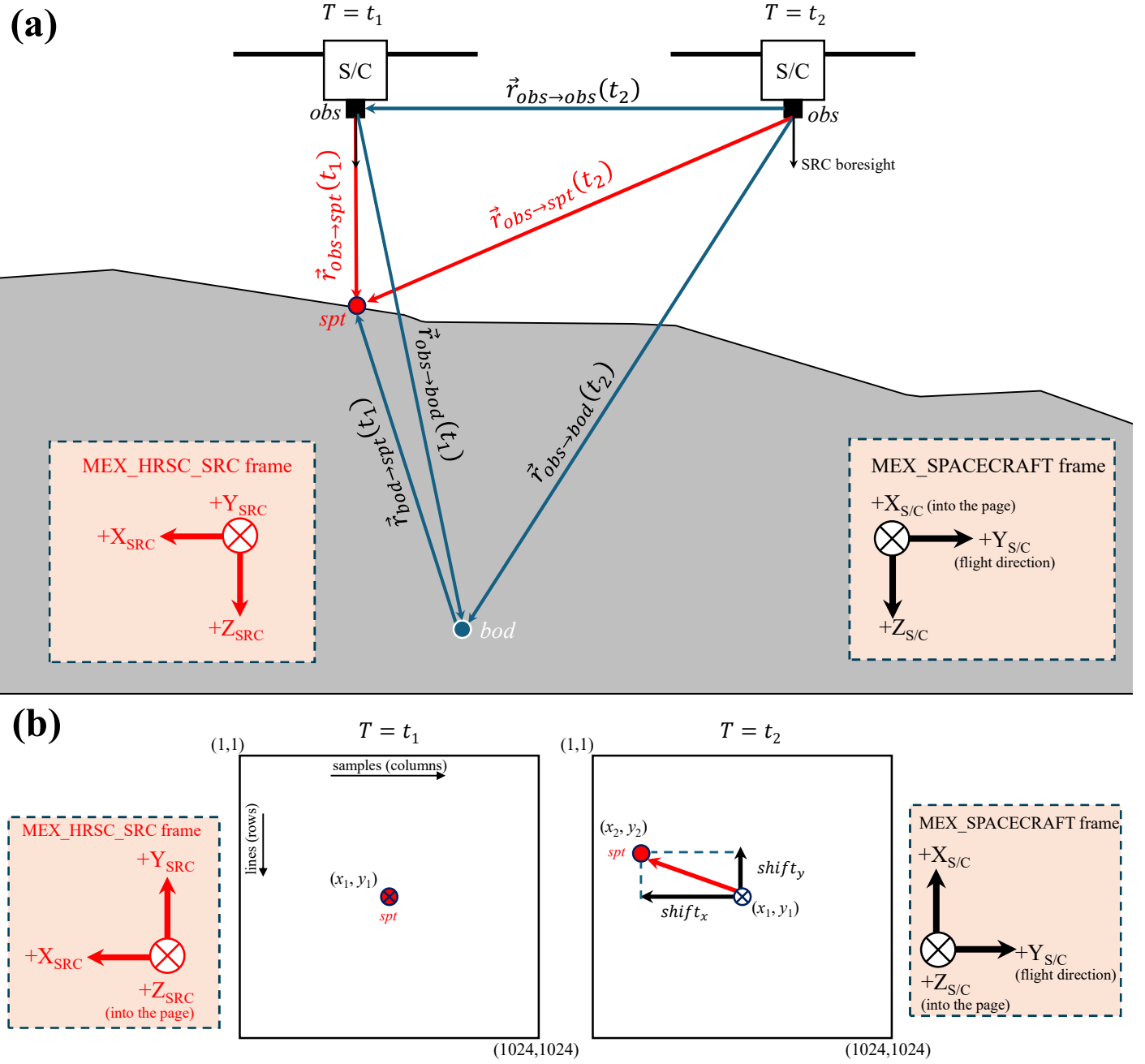


Figure S3. Schematic illustration of the SRC frame onboard the MEx spacecraft (S/C) in motion during SRC exposure. (a) spacecraft - X side view (across-track). (top) Two red arrows show vectors from observer (obs; SPICE name: "MEX\_HRSC") to the same surface-intersection point (spt) in the IAU\_PHOBOS frame. The left and right sides show the observation start time ( $t_1 = ET1'$ , or UTC1') and stop time (right side:  $t_2 = ET2'$ , or UTC2'), respectively. Not to scale. (b) SRC image plane view from SRC -Z side. The red circle on the both sides shows the spt position on the image plane before/after exposure duration. The red arrow shows the vector of a linear shift of spt whose length and orientation will be used as a component of the linear PSF of the subsequent deconvolution process. Not to scale.

### 3. Wiener deconvolution in comparison to Lucy-Richardson deconvolution

There are two major deconvolution methods to restore degraded images: Wiener deconvolution [7] and Lucy–Richardson (or Richardson–Lucy) deconvolution [8,9]. Wiener deconvolution uses a linear, closed-form filter in the frequency domain, which has achieved restoration of planetary images, such as the Near Earth Asteroid Rendezvous (NEAR) Shoemaker spacecraft Multispectral Imager (MSI) images of asteroid Eros [10,11].

Lucy–Richardson deconvolution uses an iterative, non-linear algorithm in the spatial domain, and has been applied to planetary images, including MEx/SRC images of Phobos [12] and Mars [5] (but not so high resolution and terribly blurred as those addressed in this study). Lucy–Richardson deconvolution is based on maximum-likelihood estimation assuming Poisson (photon) noise. Starting from an initial guess  $f^{(0)}$ , it refines via

$$f^{(k+1)}(x, y) = f^{(k)}(x, y) \left[ h(-x, -y) * \frac{g(x, y)}{(f^{(k)} * h)(x, y)} \right] \quad (\text{S12})$$

where “\*” denotes convolution and  $h$  is the PSF reversed. Each pass sharpens details and suppresses noise according to the Poisson likelihood.

Either would work for restoring SRC images; however, we selected Wiener deconvolution for the restoration in this study. This is because the PSF can be computed a priori from precise motion and exposure data, the closed-form Wiener filter offers a direct, non-iterative solution that (i) fully exploits the known kernel, (ii) minimizes noise amplification via the noise term, and (iii) avoids iteration-dependent artifacts and convergence issues associated with Lucy–Richardson (Table S4).

The Gaussian edge tapering affected only the outermost ~80 pixels along the image borders, while the interior region—used for geological interpretation—remained unchanged (Figure S4). Intensity profiles before and after tapering confirmed that no real surface features were degraded by this step.



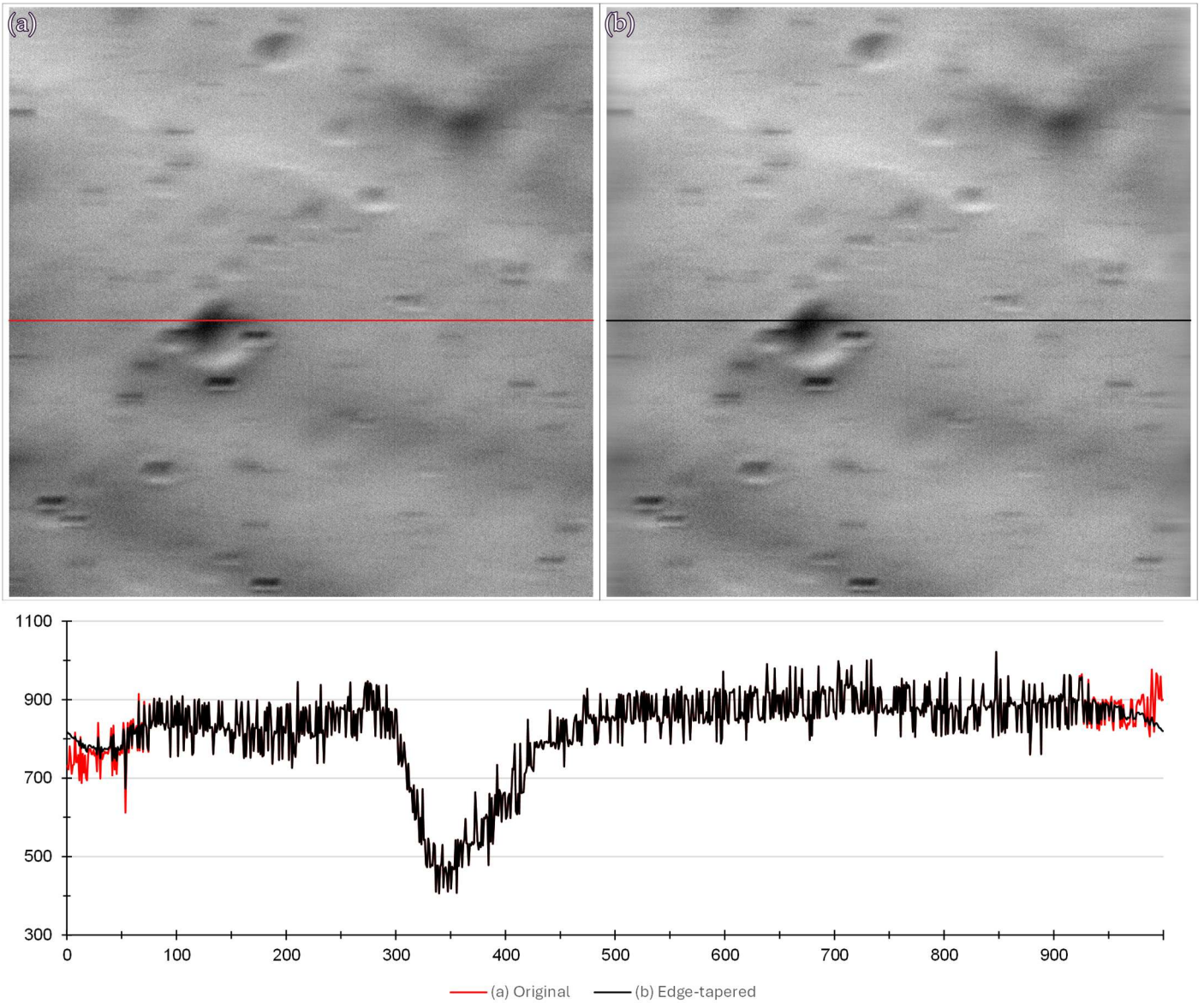


Figure S4. Comparison of typical images before and after edge tapering. (a) An image before edge tapering, (b) An image after ~81-pixel-wide edge tapering on both left and right edges.



#### 4. Geological interpretations

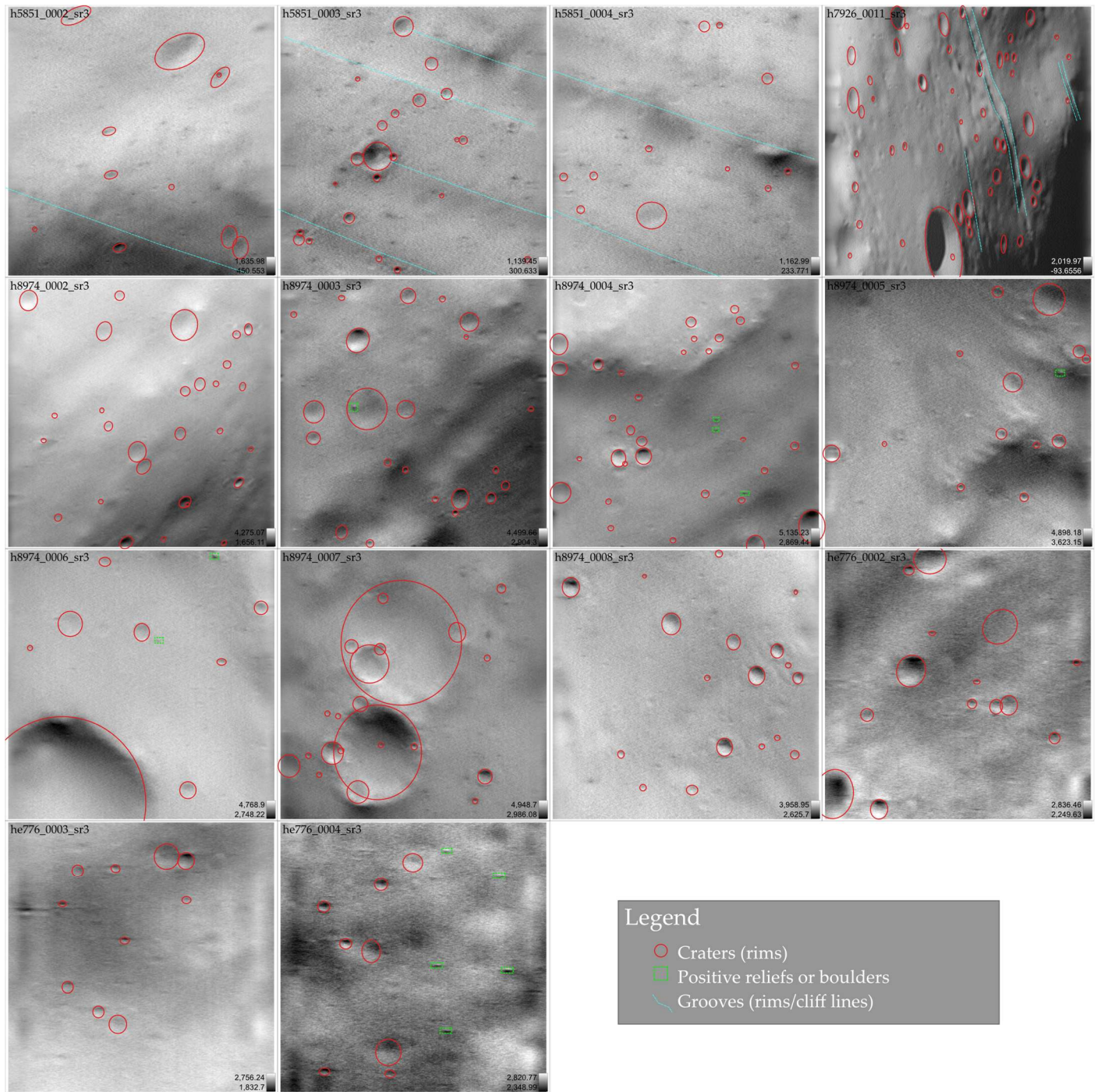


Figure S5. Geological features identified from restored SRC images. Outlines of craters, positive reliefs (likely boulders), and grooves have been manually traced by the authors. These features were indistinct in the original blurred images but become clearly visible after restoration.



## 5. Details of validation metrics in spatial domain

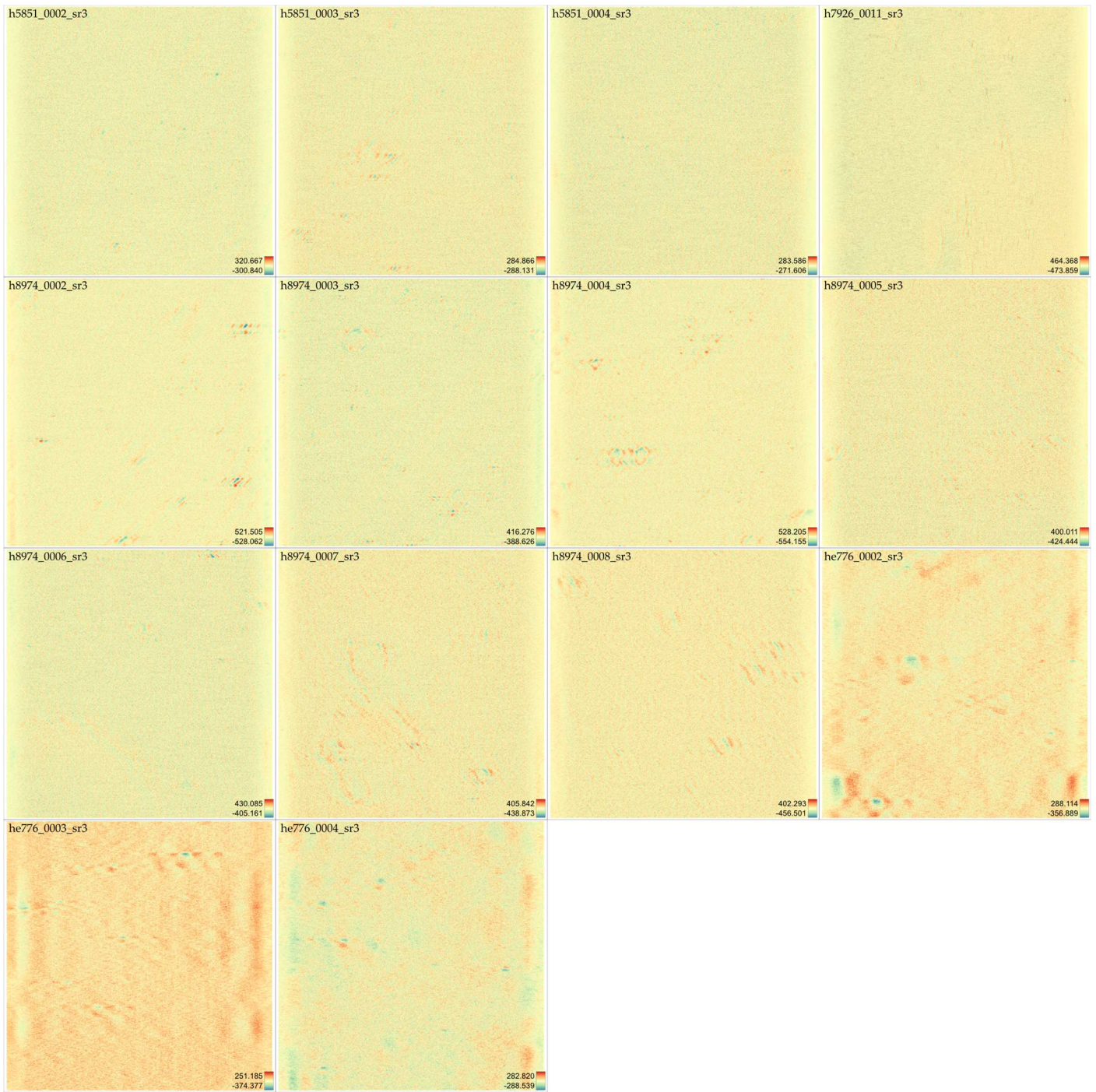


Figure S6. Signed differences  $\Delta = \hat{F} - G$ .



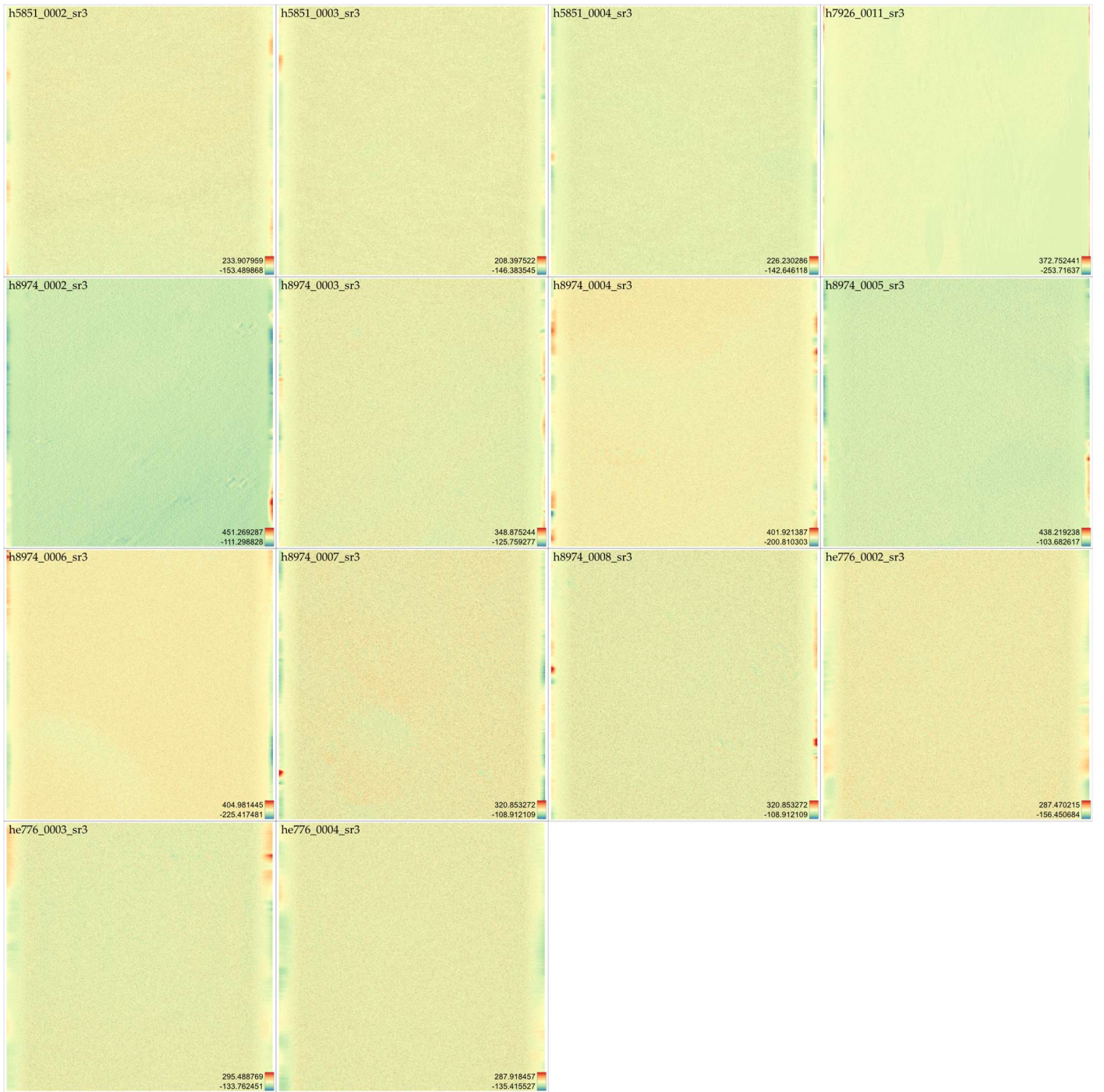


Figure S7. Forward-model residuals  $R = G - (h * \hat{F})$ . Their RMS values are reported in Table 1 of the main text.

Table S1. The list of selected MEx SRC level 3 products.

| Image ID       | Longitude (°E)* | Planetocentric latitude (°N)* | Sample resolution (meters/pixel)* | Slant distance (km)* |
|----------------|-----------------|-------------------------------|-----------------------------------|----------------------|
| h5851_0002_sr3 | -167.870        | 72.56                         | 0.875                             | 95.69                |
| h5851_0003_sr3 | 145.575         | 61.96                         | 0.858                             | 93.91                |
| h5851_0004_sr3 | 135.587         | 46.11                         | 0.856                             | 93.62                |
| h7926_0011_sr3 | 153.746         | -10.81                        | 2.671                             | 292.30               |
| h8974_0002_sr3 | -151.928        | -55.20                        | 0.931                             | 101.87               |
| h8974_0003_sr3 | -164.088        | -50.72                        | 0.923                             | 101.03               |
| h8974_0004_sr3 | -173.283        | -45.66                        | 0.917                             | 100.30               |
| h8974_0005_sr3 | 179.642         | -40.60                        | 0.915                             | 100.09               |
| h8974_0006_sr3 | 173.645         | -35.28                        | 0.914                             | 100.06               |
| h8974_0007_sr3 | 168.288         | -29.68                        | 0.917                             | 100.30               |
| h8974_0008_sr3 | 162.914         | -23.32                        | 0.924                             | 101.06               |
| he776_0002_sr3 | -29.989         | 57.82                         | 0.490                             | 53.66                |
| he776_0003_sr3 | 6.246           | 40.81                         | 0.473                             | 51.73                |
| he776_0004_sr3 | 15.872          | 31.14                         | 0.468                             | 51.19                |

\* Values were calculated at the image center of each cube by using ISIS campt application.

Table S2. The selected MEx SRC level 3 products with high-precision observation timings.

| Image ID       | UTC <sub>1</sub> ' ("START_TIME" with microsecond precision)* | UTC <sub>2</sub> ' ("STOP_TIME" with microsecond precision)* | Exposure (milliseconds) |
|----------------|---------------------------------------------------------------|--------------------------------------------------------------|-------------------------|
| h5851_0002_sr3 | 2008-07-23T04:49:59.834944                                    | 2008-07-23T04:49:59.849056                                   | 14.112                  |
| h5851_0003_sr3 | 2008-07-23T04:50:00.924944                                    | 2008-07-23T04:50:00.939056                                   | 14.112                  |
| h5851_0004_sr3 | 2008-07-23T04:50:02.014944                                    | 2008-07-23T04:50:02.029056                                   | 14.112                  |
| h7926_0011_sr3 | 2010-03-10T05:59:39.782944                                    | 2010-03-10T05:59:39.797056                                   | 14.112                  |
| h8974_0002_sr3 | 2011-01-09T14:06:28.284936                                    | 2011-01-09T14:06:28.301064                                   | 16.128                  |
| h8974_0003_sr3 | 2011-01-09T14:06:28.829936                                    | 2011-01-09T14:06:28.846064                                   | 16.128                  |
| h8974_0004_sr3 | 2011-01-09T14:06:29.374936                                    | 2011-01-09T14:06:29.391064                                   | 16.128                  |
| h8974_0005_sr3 | 2011-01-09T14:06:29.919936                                    | 2011-01-09T14:06:29.936064                                   | 16.128                  |
| h8974_0006_sr3 | 2011-01-09T14:06:30.464936                                    | 2011-01-09T14:06:30.481064                                   | 16.128                  |
| h8974_0007_sr3 | 2011-01-09T14:06:31.009936                                    | 2011-01-09T14:06:31.026064                                   | 16.128                  |
| h8974_0008_sr3 | 2011-01-09T14:06:31.554936                                    | 2011-01-09T14:06:31.571064                                   | 16.128                  |
| he776_0002_sr3 | 2015-08-26T13:18:49.613920                                    | 2015-08-26T13:18:49.634080                                   | 20.16                   |
| he776_0003_sr3 | 2015-08-26T13:18:51.248920                                    | 2015-08-26T13:18:51.269080                                   | 20.16                   |
| he776_0004_sr3 | 2015-08-26T13:18:52.090920                                    | 2015-08-26T13:18:52.111080                                   | 20.16                   |

\* A prime symbol (') shows a time including a microsecond precision.

Table S3. Results of the PSF geometries as the length and orientation of spt shifts during exposure on the SRC image plane.

| Image ID       | $shift_x$ (pixels) | $shift_y$ (pixels) | Length of PSF (pixels) <sup>1</sup> | Angle of PSF (degrees) <sup>2</sup> |
|----------------|--------------------|--------------------|-------------------------------------|-------------------------------------|
| h5851_0002_sr3 | -43.5937           | 0.2034             | 43.5942                             | 179.7327                            |
| h5851_0003_sr3 | -44.582            | 0.2079             | 44.5825                             | 179.7328                            |
| h5851_0004_sr3 | -44.8592           | 0.2016             | 44.8596                             | 179.7426                            |
| h7926_0011_sr3 | -11.3116           | 0.0071             | 11.3116                             | 179.964                             |
| h8974_0002_sr3 | 45.8297            | -0.0092            | 45.8297                             | 359.9885                            |
| h8974_0003_sr3 | 46.2751            | -0.0155            | 46.2751                             | 359.9809                            |
| h8974_0004_sr3 | 46.6702            | -0.0215            | 46.6702                             | 359.9736                            |
| h8974_0005_sr3 | 46.8149            | -0.0265            | 46.8149                             | 359.9676                            |
| h8974_0006_sr3 | 46.8688            | -0.031             | 46.8688                             | 359.9621                            |
| h8974_0007_sr3 | 46.7906            | -0.0348            | 46.7907                             | 359.9573                            |
| h8974_0008_sr3 | 46.4568            | -0.0375            | 46.4568                             | 359.9538                            |
| he776_0002_sr3 | 113.9898           | 0.5825             | 113.9913                            | 0.2928                              |
| he776_0003_sr3 | 117.8935           | 0.643              | 117.8953                            | 0.3125                              |
| he776_0004_sr3 | 118.8855           | 0.5837             | 118.8869                            | 0.2813                              |

<sup>1</sup> The value of  $(shift_x^2 + shift_y^2)^{\frac{1}{2}}$ . See the highlighted part at the bottom of Figure S3 for context.

<sup>2</sup> The value of  $\tan^{-1}(shift_y/shift_x)$  in degrees. See the highlighted part at the bottom of Figure S3 for context.

Table S4. Practical implications of the two deconvolution methods.

| Aspect           | Wiener Deconvolution                      | Lucy–Richardson Deconvolution           |
|------------------|-------------------------------------------|-----------------------------------------|
| Domain           | Frequency                                 | Spatial (iterative)                     |
| Speed            | Fast: one filter application              | Slower: many iterations                 |
| Regularization   | Built-in via noise term $S_N/S_F$         | Implicit via stopping rule (iterations) |
| Image Priors     | Requires power spectral density estimates | Implicit Poisson model + positivity     |
| Result Control   | Trade-off set by noise term               | Trade-off by number of iterations       |
| Typical Use-case | When noise is Gaussian and PSF known      | When photon-limited or low-SNR imagery  |

## Supplementary References

1. USGS-Astrogeology *Abstraction Layer for Ephemerides (ALE) 0.10.0*, 0.10.0; 2024.
2. Laura, J.R.; Mapel, J.; Hare, T. Planetary Sensor Models Interoperability Using the Community Sensor Model Specification. *Earth and Space Science* **2020**, *7*, e2019EA000713, doi:10.1029/2019EA000713.
3. Mapel, J.A.; Berry, K.; Paquette, A.; Rodriguez, K.; Stapleton, S.; Hare, T.M.; Laura, J.R. The Abstraction Layer for Ephemerides Library. In Proceedings of the 4th Planetary Data Workshop, Flagstaff, Arizona, 2019.
4. Annex, A.M.; Pearson, B.; Seignovert, B.; Carcich, B.T.; Eichhorn, H.; Mapel, J.A.; Von Forstner, J.L.F.; McAuliffe, J.; Del Rio, J.D.; Berry, K.L. SpiceyPy: a Pythonic Wrapper for the SPICE Toolkit. *Journal of Open Source Software* **2020**, *5*, 2050, doi:10.21105/joss.02050.
5. Gwinner, K.; Jaumann, R.; Hauber, E.; Hoffmann, H.; Heipke, C.; Oberst, J.; Neukum, G.; Ansan, V.; Bostelmann, J.; Dumke, A.; et al. The High Resolution Stereo Camera (HRSC) of Mars Express and its approach to science analysis and mapping for Mars and its satellites. *Planetary and Space Science* **2016**, *126*, 93–138, doi:10.1016/j.pss.2016.02.014.
6. Neukum, G.; Jaumann, R. HRSC: The High Resolution Stereo Camera of Mars Express. In *Mars Express: The Scientific Payload*; ESA: 2004; Volume SP-1240, pp. 17–35.
7. Wiener, N. *Extrapolation, Interpolation, and Smoothing of Stationary Time Series: With Engineering Applications*; The MIT Press: 1949.
8. Richardson, W.H. Bayesian-Based Iterative Method of Image Restoration. *J. Opt. Soc. Am.* **1972**, *62*, 55–59, doi:10.1364/JOSA.62.000055.
9. Lucy, L.B. An iterative technique for the rectification of observed distributions. *The Astronomical Journal* **1974**, *79*, 745–754, doi:10.1086/111605.
10. Li, H.; Robinson, M.S.; Murchie, S. Preliminary Remediation of Scattered Light in NEAR MSI Images. *Icarus* **2002**, *155*, 244–252, doi:10.1006/icar.2001.6745.
11. Golish, D.R.; DellaGiustina, D.N.; Becker, K.J.; Bennett, C.A.; Robinson, M.; Crombie, M.K. Blur remediation in NEAR MSI images. *Icarus* **2023**, *400*, 115536, doi:10.1016/j.icarus.2023.115536.
12. Pasewaldt, A.; Oberst, J.; Willner, K.; Beisembin, B.; Hoffmann, H.; Matz, K.D.; Roatsch, T.; Michael, G.; Cardesín-Moinelo, A.; Zubarev, A.E. Astrometric observations of Phobos with the SRC on Mars Express. *Astronomy & Astrophysics* **2015**, *580*, doi:10.1051/0004-6361/201525957.

**Disclaimer/Publisher’s Note:** The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.